Cells Test - Answer Key

1. Describe the process of cellular respiration, including the three main stages and the role of the mitochondria.

Answer: Cellular respiration is the process by which cells convert nutrients, primarily glucose, into usable energy in the form of ATP. The three main stages of cellular respiration are glycolysis, the Krebs cycle, and the electron transport chain. Glycolysis occurs in the cytoplasm and does not require oxygen, producing a small amount of ATP and NADH. The Krebs cycle occurs in the mitochondrial matrix and produces more ATP, NADH, and $FADH_2$. The electron transport chain occurs in the inner mitochondrial membrane and produces most of the cell's ATP through chemiosmosis, requiring oxygen as the final electron acceptor. The mitochondria play a crucial role in cellular respiration, generating most of the cell's ATP through the Krebs cycle and electron transport chain.

Explanation: This question requires a detailed description of the process of cellular respiration, including the three main stages and the role of the mitochondria.